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Divide and Conquer – Searching & Basic Problems

Binary Search, Finding Maximum and Minimum using Divide and Conquer

Dr. U. Samson Ebenezer

Question 1: Search an Element in a Sorted Array

Given a sorted array of integers and a key value, use the Binary Search (Divide and Conquer) method to determine whether the key is present in the array. If found, print its position; otherwise, print "Element not found".

Test Case 1

Input

5
10 20 30 40 50
30

Expected Output

Element found at position 3

Test Case 2

Input

5
10 20 30 40 50
25

Expected Output

Element not found

Question 2: Find the Index of a Given Number

Given a sorted array and a target number, find the index of the target using Binary Search. Assume array indexing starts from 0.

Test Case 1

Input

6
2 4 6 8 10 12
8

Expected Output

Index = 3

Test Case 2

Input

6
2 4 6 8 10 12
5

Expected Output

Index = -1

Question 3: Count the Number of Iterations in Binary Search

Given a sorted array and a key, perform Binary Search and display the number of iterations required to find the element.

Test Case 1

Input

7
5 10 15 20 25 30 35
25

Expected Output

Element found

Iterations = 2

Test Case 2

Input

7
5 10 15 20 25 30 35
40

Expected Output

Element not found

Iterations = 3

Question 4: Find the First Occurrence of an Element

Given a sorted array containing duplicate elements, find the first occurrence of a given key using Binary Search.

Test Case 1

Input

8
1 2 2 2 3 4 5 6
2

Expected Output

First occurrence at index 1

Test Case 2

Input

8
1 2 2 2 3 4 5 6
7

Expected Output

Element not found

Question 5: Find the Last Occurrence of an Element

Given a sorted array containing duplicate elements, find the last occurrence of a given key using Binary Search.

Test Case 1

Input

8
1 2 2 2 3 4 5 6
2

Expected Output

Last occurrence at index 3

Test Case 2

Input

8
1 2 2 2 3 4 5 6
9

Expected Output

Element not found

Question 6: Highest and Lowest Temperature Recorded

A weather station records temperatures over a week. Find the **minimum** and **maximum** temperature using the Divide and Conquer approach.

Input

7

32 28 35 25 31 29 37

Expected Output

Minimum Temperature = 25

Maximum Temperature = 37

Question 6: Best and Worst Student Marks

A teacher wants to identify the highest and lowest marks scored in a class using Divide and Conquer.

Input

8

78 92 65 88 95 72 81 69

Expected Output

Minimum Mark = 65

Maximum Mark = 95

Question 8: Stock Price Analysis

An investor wants to know the minimum and maximum stock prices recorded during a trading week.

Input

6

450 520 480 610 430 590

Expected Output

Minimum Stock Price = 430

Maximum Stock Price = 610

Question 9: Fastest and Slowest Race Times

A sports analyst needs to find the fastest (minimum) and slowest (maximum) race completion times.

Input

9

14 12 18 11 15 13 17 16 10

Expected Output

Fastest Time = 10

Slowest Time = 18

Question 10: Largest and Smallest Building Heights

An urban planner is analyzing building heights in a city block and wants to determine the tallest and shortest buildings.

Input

10

120 150 98 175 140 165 110 190 130 145

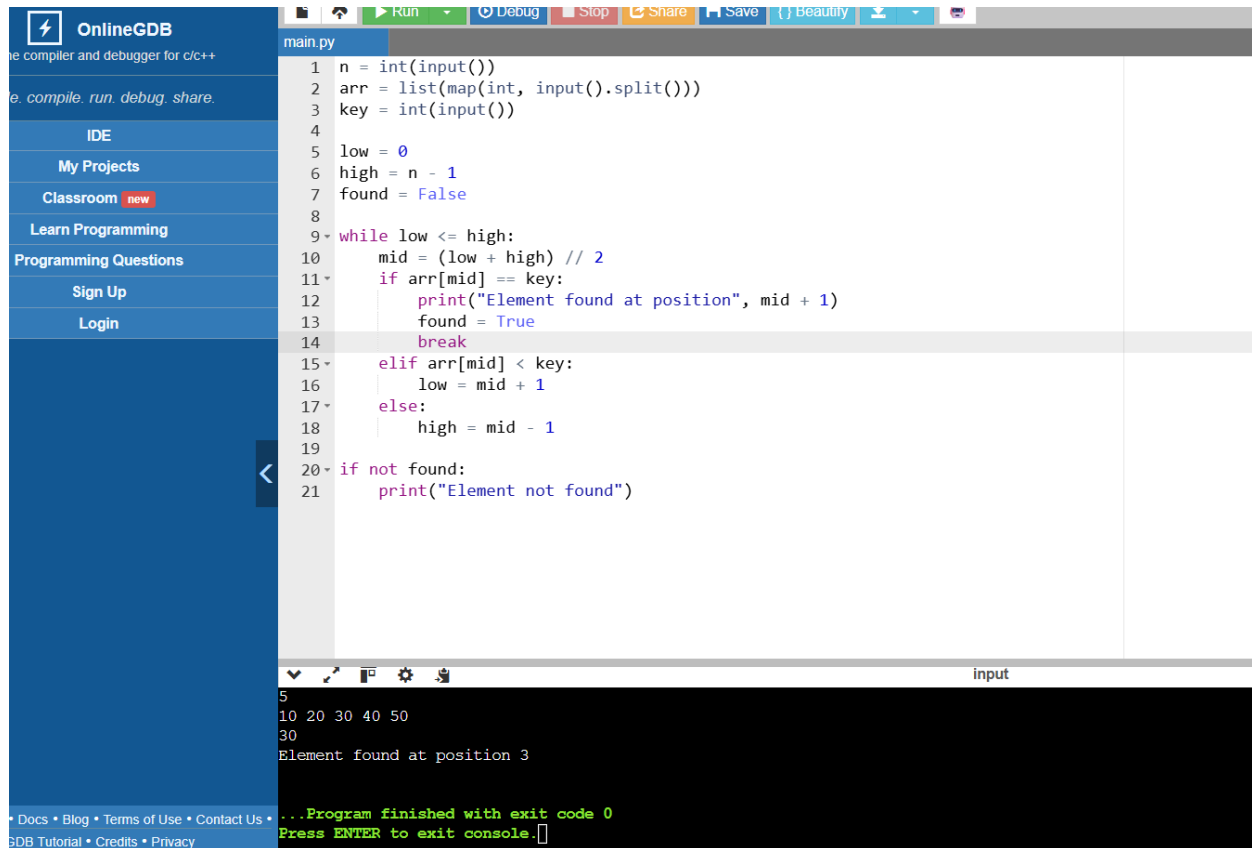
Expected Output

Minimum Height = 98

Maximum Height = 190

OUTPUTS:

Question 1: Search an Element in a Sorted Array



The screenshot displays the OnlineGDB web interface. On the left is a blue sidebar with navigation links: IDE, My Projects, Classroom (marked 'new'), Learn Programming, Programming Questions, Sign Up, and Login. The main area shows a Python file named 'main.py' with a binary search algorithm. The code takes an integer 'n' and a space-separated string of integers, converts them to a list, and searches for a 'key' using a while loop. The console at the bottom shows the input '10 20 30 40 50' and '30', resulting in the output 'Element found at position 3'. The program ends with 'Program finished with exit code 0'.

```
1 n = int(input())
2 arr = list(map(int, input().split()))
3 key = int(input())
4
5 low = 0
6 high = n - 1
7 found = False
8
9 while low <= high:
10     mid = (low + high) // 2
11     if arr[mid] == key:
12         print("Element found at position", mid + 1)
13         found = True
14         break
15     elif arr[mid] < key:
16         low = mid + 1
17     else:
18         high = mid - 1
19
20 if not found:
21     print("Element not found")
```

input

5
10 20 30 40 50
30
Element found at position 3

...Program finished with exit code 0
Press ENTER to exit console.

Question 2: Find the Index of a Given Number

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main.py

```
1 n = int(input())
2 arr = list(map(int, input().split()))
3 target = int(input())
4
5 low = 0
6 high = n - 1
7 index = -1
8
9 while low <= high:
10     mid = (low + high) // 2
11     if arr[mid] == target:
12         index = mid
13         break
14     elif arr[mid] < target:
15         low = mid + 1
16     else:
17         high = mid - 1
18
19 print("Index =", index)
```



input

```
6
2 4 6 8 10 12
8
Index = 3

...Program finished with exit code 0
Press ENTER to exit console.
```

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Question 3: Count the Number of Iterations in Binary Search

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main.py

```
1 n = int(input())
2 arr = list(map(int, input().split()))
3 key = int(input())
4
5 low = 0
6 high = n - 1
7 iterations = 0
8 found = False
9
10 while low <= high:
11     iterations += 1
12     mid = (low + high) // 2
13
14     if arr[mid] == key:
15         found = True
16         break
17     elif arr[mid] < key:
18         low = mid + 1
19     else:
20         high = mid - 1
21
22 if found:
23     print("Element found Iterations =", iterations)
24 else:
25     print("Element not found Iterations =", iterations)
```



input

```
7
5 10 15 20 25 30 35
40
Element not found Iterations = 3

...Program finished with exit code 0
Press ENTER to exit console.
```

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Question 4: Find the First Occurrence of an Element

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main.py

```
1 n = int(input())
2 arr = list(map(int, input().split()))
3 key = int(input())
4
5 low = 0
6 high = n - 1
7 result = -1
8
9 while low <= high:
10     mid = (low + high) // 2
11
12     if arr[mid] == key:
13         result = mid
14         high = mid - 1
15     elif arr[mid] < key:
16         low = mid + 1
17     else:
18         high = mid - 1
19
20 if result != -1:
21     print("First occurrence at index", result)
22 else:
23     print("Element not found")
```



8
1 2 2 2 3 4 5 6
2
First occurrence at index 1

...Program finished with exit code 0
Press ENTER to exit console.

Question 5: Find the Last Occurrence of an Element

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main.py

```
1 n = int(input())
2 arr = list(map(int, input().split()))
3 key = int(input())
4
5 low = 0
6 high = n - 1
7 result = -1
8
9 while low <= high:
10     mid = (low + high) // 2
11
12     if arr[mid] == key:
13         result = mid
14         low = mid + 1
15     elif arr[mid] < key:
16         low = mid + 1
17     else:
18         high = mid - 1
19
20 if result != -1:
21     print("Last occurrence at index", result)
22 else:
23     print("Element not found")
```

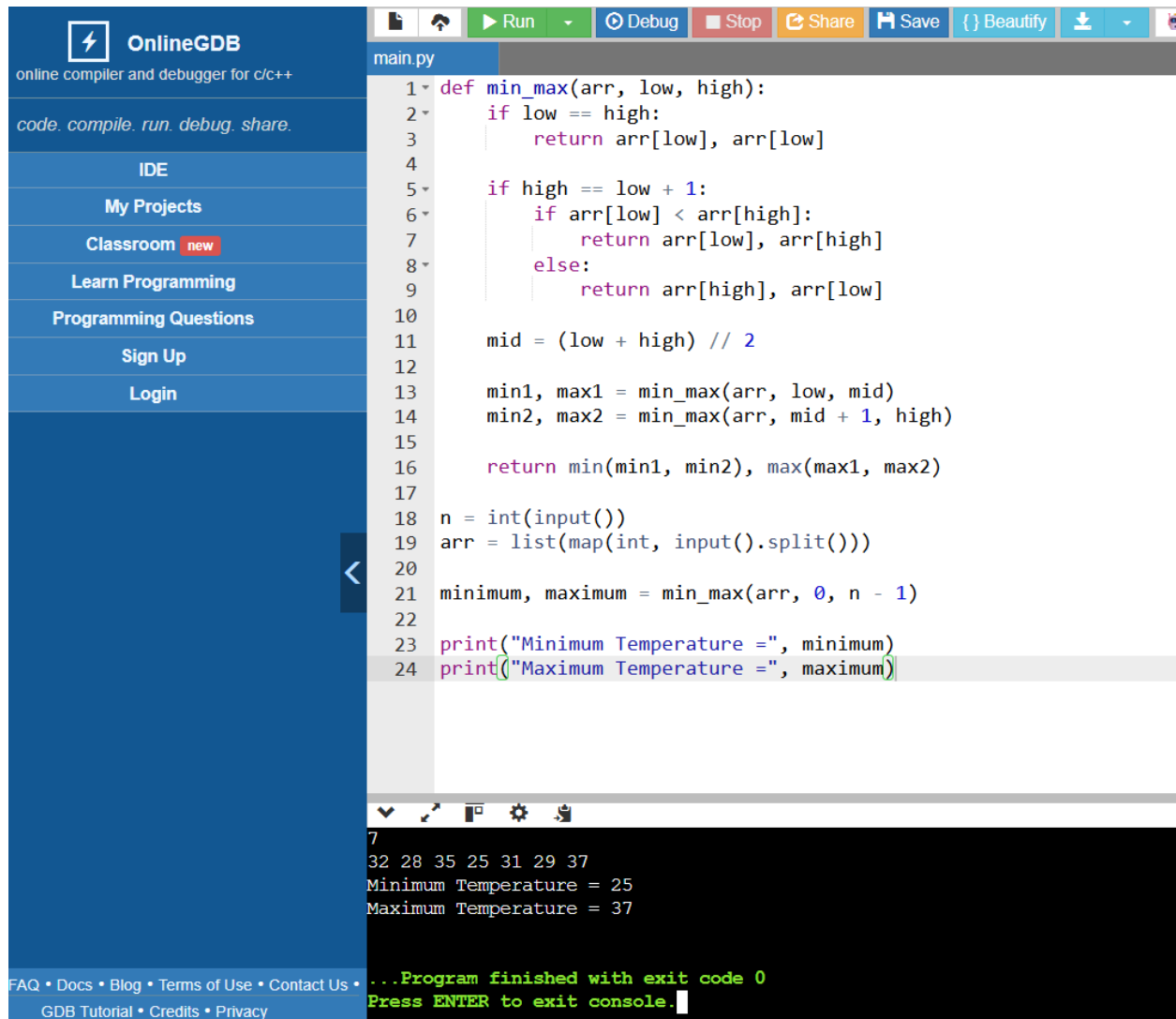
      

```
8
1 2 2 2 3 4 5 6
2
Last occurrence at index 3

...Program finished with exit code 0
Press ENTER to exit console.
```

Question 6: Highest and Lowest Temperature Recorded



The screenshot shows the OnlineGDB IDE interface. On the left is a sidebar with navigation links: OnlineGDB, code, compile, run, debug, share, IDE, My Projects, Classroom (new), Learn Programming, Programming Questions, Sign Up, and Login. The top toolbar contains icons for Run, Debug, Stop, Share, Save, Beautify, and a download icon. The main editor displays a Python file named 'main.py' with the following code:

```
1 def min_max(arr, low, high):
2     if low == high:
3         return arr[low], arr[low]
4
5     if high == low + 1:
6         if arr[low] < arr[high]:
7             return arr[low], arr[high]
8         else:
9             return arr[high], arr[low]
10
11     mid = (low + high) // 2
12
13     min1, max1 = min_max(arr, low, mid)
14     min2, max2 = min_max(arr, mid + 1, high)
15
16     return min(min1, min2), max(max1, max2)
17
18 n = int(input())
19 arr = list(map(int, input().split()))
20
21 minimum, maximum = min_max(arr, 0, n - 1)
22
23 print("Minimum Temperature =", minimum)
24 print("Maximum Temperature =", maximum)
```

Below the editor is a console window showing the program's output:

```
7
32 28 35 25 31 29 37
Minimum Temperature = 25
Maximum Temperature = 37
...Program finished with exit code 0
Press ENTER to exit console.
```

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Question 7: Best and Worst Student Marks

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main.py

```
1 def min_max(arr, low, high):
2     if low == high:
3         return arr[low], arr[low]
4
5     if high == low + 1:
6         if arr[low] < arr[high]:
7             return arr[low], arr[high]
8         else:
9             return arr[high], arr[low]
10
11     mid = (low + high) // 2
12
13     min1, max1 = min_max(arr, low, mid)
14     min2, max2 = min_max(arr, mid + 1, high)
15
16     return min(min1, min2), max(max1, max2)
17
18 n = int(input())
19 arr = list(map(int, input().split()))
20
21 minimum, maximum = min_max(arr, 0, n - 1)
22
23 print("Minimum Mark =", minimum)
24 print("Maximum Mark =", maximum)
```





```
8
78 92 65 88 95 72 81 69
Minimum Mark = 65
Maximum Mark = 95
...Program finished with exit code 0
Press ENTER to exit console.
```

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Question 8: Stock Price Analysis

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main.py

```
1 def min_max(arr, low, high):
2     if low == high:
3         return arr[low], arr[low]
4
5     if high == low + 1:
6         if arr[low] < arr[high]:
7             return arr[low], arr[high]
8         else:
9             return arr[high], arr[low]
10
11     mid = (low + high) // 2
12
13     min1, max1 = min_max(arr, low, mid)
14     min2, max2 = min_max(arr, mid + 1, high)
15
16     return min(min1, min2), max(max1, max2)
17
18 n = int(input())
19 arr = list(map(int, input().split()))
20
21 minimum, maximum = min_max(arr, 0, n - 1)
22
23 print("Minimum Stock Price =", minimum)
24 print("Maximum Stock Price =", maximum)
```

10

120 150 98 175 140 165 110 190 130 145

Minimum Stock Price = 98

Maximum Stock Price = 190

...Program finished with exit code 0

Press ENTER to exit console.

Question 9: Fastest and Slowest Race Times

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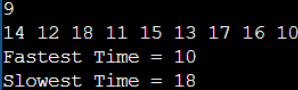
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main.py

```
1 def min_max(arr, low, high):
2     if low == high:
3         return arr[low], arr[low]
4
5     if high == low + 1:
6         if arr[low] < arr[high]:
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8         else:
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11     mid = (low + high) // 2
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13     min1, max1 = min_max(arr, low, mid)
14     min2, max2 = min_max(arr, mid + 1, high)
15
16     return min(min1, min2), max(max1, max2)
17
18 n = int(input())
19 arr = list(map(int, input().split()))
20
21 minimum, maximum = min_max(arr, 0, n - 1)
22
23 print("Fastest Time =", minimum)
24 print("Slowest Time =", maximum)
```


9
14 12 18 11 15 13 17 16 10
Fastest Time = 10
Slowest Time = 18
...Program finished with exit code 0
Press ENTER to exit console.

Question 10: Largest and Smallest Building Heights

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4
5     if high == low + 1:
6         if arr[low] < arr[high]:
7             return arr[low], arr[high]
8         else:
9             return arr[high], arr[low]
10
11     mid = (low + high) // 2
12
13     min1, max1 = min_max(arr, low, mid)
14     min2, max2 = min_max(arr, mid + 1, high)
15
16     return min(min1, min2), max(max1, max2)
17
18 n = int(input())
19 arr = list(map(int, input().split()))
20
21 minimum, maximum = min_max(arr, 0, n - 1)
22
23 print("Minimum Height =", minimum)
24 print("Maximum Height =", maximum)
```

```
6
450 520 480 610 430 590
Minimum Height = 430
Maximum Height = 610
...Program finished with exit code 0
Press ENTER to exit console.
```